



SL9 SLAM RTK User Manual

Manual Revision

SatLab GEOSOLUTIONS AB

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SL9

SLAM RTK

User Manual



Preface

Introduction

Welcome to the SatLab SL9 receiver. This introduction describes how to use this product.

Experience requirement

To help you use the SatLab products better, SatLab suggests that you read the instructions carefully. If you are unfamiliar with the products, please refer to <https://www.satlab.com.se/>

Tips for safe use



Notice: These are special operations and need your special attention. Please read them carefully.



Warning: The contents here are very important as the wrong operation may damage the machine. This can lead to the loss of data, or break the system and endanger your safety.

Exclusions

Before using the product, please read these operating instructions carefully, as they will help you to use it better. GeoSolution i Göteborg AB assumes no responsibility if you fail to operate the product according to the instructions or operate it wrongly because you have misunderstood them.

SatLab is committed to constantly perfecting the product's functions and performance, improving its service quality and reserves the right to change these operating instructions without notice.

We have checked the contents of the instructions and software and hardware but please note the possibility of deviation. The pictures in the operating instructions are for reference only and the actual products may vary from them.

Technology and service

If you have any technical issues, please call the SatLab technology department for help, and we will answer your question.

Relevant information

You can obtain this introduction by:

Purchasing SatLab products: this manual is found in the instrument container and will help you to operate the instrument.

Logging on to the SatLab official website and downloading the electronic version of this introduction from Partner Center: <http://members.satlab.com.se/>

Advice

If you have any suggestions for this product, please email us: info@satlab.com.se. Your feedback information will help us to improve our product and service.

Contents

Preface	4
Introduction	4
Experience requirement.....	4
Tips for safe use	4
Exclusions.....	4
Technology and service	4
Relevant information.....	4
Advice.....	5
Contents.....	6
Chapter 1	9
Overview	9
1.1 Foreword.....	10
1.2 Features.....	10
1.3 Use and precautions.....	10
Chapter 2	13
Product introduction	13
2.1 Overall appearance	14
2.1.1 Structure	14
2.2 Basic functions	15
2.2.1 Battery installation	15
2.2.2 Battery disassembly.....	16
2.2.3 Key function.....	17
2.2.4 LED indicator function	17
2.2.5 Data copies	18
2.2.6 Battery charging	18
2.2.7 Firmware upgrade	18
2.2.8 Precautions for use.....	19
2.3 APP software operation.....	19
Chapter 3	22

Equipment	22
3.1 Device connection	23
3.1.1 Register	23
3.1.2 NFC connection.....	24
3.2 Base	24
3.2.1 Template.....	24
3.2.2 Set manually	25
3.2.3 Station location.....	26
3.3 Rover.....	30
3.3.1 Template.....	30
3.3.2 Set manually	31
3.3.3 Rover station data link.....	31
3.3.4 Other options for rover	34
3.4 Additional function.....	35
3.4.1 Module information	35
3.4.2 Registration	36
3.4.3 Satellite tracking switch.....	36
3.4.4 Receiver settings.....	36
3.4.5 Service information	37
3.4.6 Restore System	38
3.4.7 Network Diagnosis.....	38
3.4.8 Data Transmission	38
3.5 Static collection	39
3.5.1 Static collection Settings.....	39
3.5.2 Static file management.....	40
3.6 Device information	41
3.7 Data debugging.....	42
3.8 Device check	43
Chapter 4	44
Measure.....	44
4.1 Detail survey	45
4.1.1 Manual collection	45

4.1.2 Automatic acquisition.....	46
4.1.3 Average collection	47
4.1.4 PPK collection	49
4.2 General Measurement	50
4.2.1 Tilt measurement	50
4.2.2 SLAM-RTK mode	52
4.2.3 SLAM image measurement mode	53
4.3 SLAM.....	56
4.3.1 Task management	56
4.3.2 Project management.....	63
4.4 Download job data	66
4.5 Data processing	68
4.5.1 New construction.....	68
4.5.2 Configure the coordinate system	69
4.5.3 Point cloud solution	70
4.5.4 Calculation of cutting and filling	71

Chapter 1

Overview

This chapter contains:

- Foreword
- Features
- Use and precautions

1.1 Foreword

SL9 SLAM RTK is a high-integration, high-precision medium-range handheld 3D real-world reconstruction device that supports real-time preview and data collection. It features a high-performance laser lens, camera module, GNSS positioning board, antenna, MEMS sensor, long-lasting battery, and multiple communication modules. The device comes with an easy-to-use APP, making it quick to get started with measurements.

1.2 Features

1. Real-time RTK fusion function, no post-processing;
2. Real-time RTK measurement function;
3. Strong endurance.

1.3 Use and precautions

The SL9 SLAM RTK is designed to be resistant to both chemicals and impact, but precision instruments require careful use and maintenance and care should be taken.

Notice:

This equipment complies with radiation exposure limits set forth for a controlled environment. This equipment should be installed and operated with a minimum distance of 1m between the radiator and your body. This transmitter must not be co-located or operated in conjunction with any other antenna or transmitter.



We recommend that a distance of 0.31m or more be maintained between this device and people during operation. Operations closer than this distance is not recommended.

The product shall not be permanently installed outdoors because protection against water and dust is not guaranteed Please bring back to indoors in rainy weather.

This device needs a license from the national regulatory authority before use.

The receiver must be within the specified temperature range when it is used and stored. For detailed requirements, please refer to Chapter 3: Technical specification.

Notice:

The user is cautioned that changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

(1) this device may not cause harmful interference.

(2) this device must accept any interference received, including interference that may cause undesired operation.



This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation.

If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
 - Increase the separation between the equipment and receiver.
 - Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
 - Consult the dealer or an experienced radio/TV technician for help.
-



Notice: The user manual for LE-LAN devices shall contain instructions related to the restrictions mentioned in the above sections, namely that:

- i. the device for operation in the band 5150–5250 MHz is only for indoor use to reduce the potential for harmful interference to co-channel mobile satellite systems;
 - ii. for devices with detachable antenna(s), the maximum antenna gain permitted for devices in the bands 5250-5350 MHz and 5470-5725 MHz shall be such that the equipment still complies with the e.i.r.p. limit;
 - iii. for devices with detachable antenna(s), the maximum antenna gain permitted for devices in the band 5725-5850 MHz shall be such that the equipment still complies with the e.i.r.p. limits as appropriate; and
 - iv. where applicable, antenna type(s), antenna models(s), and worst-case tilt angle(s) necessary to remain compliant with the e.i.r.p. elevation mask requirement set forth in section 6.2.2.3 shall be clearly indicated.
-



Notice:

AUTION RISK OF EXPLOSION IF BATTERY IS REPLACED BY AN INCORRECT TYPE. DISPOSE OF USED BATTERIES ACCORDING TO THE INSTRUCTIONS

Do not disposal of a battery into fire or a hot oven, or mechanically crushing or cutting of a battery, that can result in an explosion;

Do not leaving a battery in an extremely high temperature surrounding environment that can result in an explosion or the leakage of flammable liquid or gas

Do not expose the battery to extremely low air pressure that may result in an explosion or the leakage of flammable liquid or gas



Notice: The device has requirements for authorization of use exist in the Union on putting into service.

To ensure the continuous tracking observation of the satellite and quality of the satellite signal, the space above the station should be as wide as possible, with no obstacles above the 15° elevation angle. To reduce the interference of various electromagnetic waves on the GNSS satellite signal, there should be no strong electromagnetic interference within a range of about 200 meters around the station, such as TV towers, microwave stations, and high-voltage transmission lines. To avoid or reduce the occurrence of multipath effects, the station should be away from terrain and features with strong reflectors, such as high-rise buildings, water, etc.

Chapter 2

Product introduction

This chapter contains:

- Overall appearance
- Basic functions
- Software operation
- Data processing

2.1 Overall appearance

The appearance of this product is mainly divided into two parts: the main engine and the battery bar.



Figure 2-1-1

Schematic diagram in three dimensions

1-Host 2-Battery bar

2.1.1 Structure



Figure 2-1-2

1-Screen 2-Mode light 3-Precision light/Track light 4-Signal light 5-Camera 6-LiDar 7-SIM card interface 8-Power button 9-Type-C interface 10-External antenna interface 11-NFC
12-Battery bar button 13-Power display 14-Battery bar charging interface

- Camera component: Obtain the left, front and right view photo data for point cloud coloring.
- Accuracy warning light: used for preliminary evaluation of user operation accuracy.
- Power switch button: the function of turning on and off the host.
- External antenna interface: used for external radio antenna.
- NFC interface: used to control the App to quickly establish a connection with the host through NFC.
- Battery bar charging port: used for battery bar charging.
- Battery bar power button: the initial display of the remaining battery power.
- External SIM card interface: used for external SIM card, providing network data traffic.
- Type-C data copy interface: used for data copy, including point cloud data download and firmware copy upgrade.

**Notice:**

When you do not need to plug or unplug the SIM card and Type-C data cable temporarily, please cover it to achieve the purpose of waterproof and dustproof.

2.2 Basic functions

2.2.1 Battery installation

1. Place the device on a stable plane, open the battery lock lever and press the safety button;
2. Insert the battery along the guide slot into the bottom of the device and ensure that it is inserted in place;
3. Press the locking lever to lock the battery and complete the installation.



Figure 2-2-1



Notice:

The battery needs to be locked before it starts working. Failure to lock the battery may cause the device to fall or abnormal power consumption.

2.2.2 Battery disassembly

1. Place the device on a stable plane, do not disassemble in the air; open the battery locking lever and press the safety button;
2. Push the battery out along the guide slot and separate it from the host unit;
3. After the battery and the host unit are completely separated, turn back the locking lever then complete the disassembly.



Figure 2-2-2



Notice:

The battery cannot be removed in the air. The customer shall bear the damage caused by the device removed in the air.

2.2.3 Key function

Function	Key operation	Device status
Power on	Short press 1 second	The three lights remain green to stay on, and when the tracking satellite and signal lights, switch from green to off, the device enters standby mode.
Power off	Hold down for 6 seconds	When the device is in standby mode, press and hold the button for 6 seconds, the accuracy and signal lights will light up in yellow, release the button and the lights will go out to indicate that the shutdown is complete.

2.2.4 LED indicator function

Type of LED lamp	Indicate status	Meaning
Mode light	Green	The device is in RTK mode
	Blue	The device is in SLAM mode
Precision lights	Red	Indicates that poor accuracy is not available. Accuracy greater than 10cm error
	Yellow	Indicates general accuracy, accuracy in 6-10cm error
	Green	Indicates good accuracy, within 5cm error range
	Off	Indicates that acquisition/work has not started
Signal lamp	Green	Blinking green when receiving RTCM
	Off	RTCM not received

2.2.5 Data copies

- Use the USB cable in the kit. When the device is shut off, connect it to the USB 3.0 port of the computer, the computer will identify the static disc drive, enter the disc drive and select the corresponding project file to copy.



Figure 2-2-3



Figure 2-2-4

2.2.6 Battery charging

Charge by using the adapter included in the kit. The battery LED blinks to indicate that it is charging, the charging time is approximately 2 hours, and the indicator light during charging shows the current power level.



Notice:

Battery charging is recommended to charge a single battery or charge the device when it is off. Otherwise, the charging may not be completed.

2.2.7 Firmware upgrade

1. Use the USB cable in the kit. When the device is shut off, connect it to the USB port of the computer, the computer detects the update disc drive, copy the upgraded firmware package to the update disc drive, and restart the device then complete the upgrade.

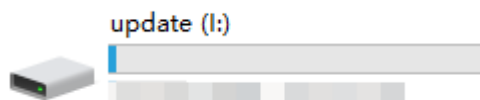


Figure 2-2-5

2. Configure remote firmware upgrade via a remote server. The device automatically detects the firmware during the boot-up process and then selects the upgrade.


Notice:

1. Do not force power off during the upgrade process. If forced power off, the instrument may be damaged and abnormal.
2. Do not upgrade the firmware when the battery level is lower than 15%

2.2.8 Precautions for use

1. SLAM RTK is a precision mapping device, which may be damaged by falling or external collision, resulting in abnormal operation or inaccurate precision.
2. To ensure that SLAM RTK is enabled, do not block the radar and camera field of view, otherwise it may lead to map building failure or accuracy decline.
3. Using the base of SLAM RTK can effectively ensure the initialization accuracy of the device; uneven ground will cause the device to shake, which may lead to the failure of initialization or increase the thickness of map building.
4. When using the device, try to avoid rapid rotation or large shaking of the device. Large shaking may lead to failure of map building or decrease of map accuracy.
5. The waterproof level of SLAM RTK is IP54, so do not use it in the environment that exceeds this protection level. It is recommended to clean the device with a soft dry cloth or built-in cleaning cloth. Keep the radar and lens parts clean and do not touch them directly with your hands.
6. Do not cover the heat dissipation area during use. If the heat dissipation area is seriously blocked, the heat dissipation efficiency will be greatly reduced and the device will automatically shut down.

2.3 APP software operation

Hi-Survey is a controller software designed to work with Hi-Target SLAM RTK products. After connecting the device, users can view various information, switch between different working modes, and download specific data. From the user's perspective, the software aims to provide more precise measurements, more user-friendly operations, and a wider range of functions to support diverse measurement tasks.

2.3.1 Precautions for operation

1. When connecting the device with the software of this product, if the controller device is disconnected from the host during the operation process due to environmental or other factors, you can wait for the controller device to automatically reconnect or manually connect. After reconnection, the trajectory before the disconnection and the point cloud collected after the recovery will be restored. The data will be stored in the device throughout the process.

2. When initializing, the device should be oriented towards buildings and other features, avoid moving vehicles and crowds, and open fields; ensure that the LiDar is not blocked, and avoid a group of people surrounding the device.
3. If there are people accompanying during the collection process, please let them follow behind the collector, and it is not recommended to stand side by side.
4. Avoid violent rotation of the measuring rod (body) or violent shaking, and keep stable movement during the collection process; after placing it steadily, click the collection 1-2s later.
5. When entering or leaving a room, open the door in advance, turn your back and move slowly.
6. Please take the stairs to go up and down the floor. Taking the elevator may lead to the failure of building map. When going up and down the stairs, avoid sticking to the wall.
7. In the corridor, the device should be tilted forward appropriately to align the laser with the end of the corridor to avoid drift in the forward direction.
8. If the main power quantity reaches or falls below 5% during the operation, the current task will be forcibly terminated and data will be saved automatically.
9. Before operation, please check whether the battery bar is installed in place and whether the safety bolt is locked. Check whether the radar is dirty. If so, wipe it clean.
10. Before operation, the measurement route should be planned according to the survey area, and the power of the device and the power of the hand book should be checked to avoid missing or insufficient power and need to be reworked.
11. Check whether the camera and laser head functions are normal before operation.
12. If the measurement area is an indoor scene, open all doors first if conditions permit.
13. If vehicles pass through the collection process, the device can be turned away from the moving object and then collected after the moving object passes through.
14. When turning, increase the radius of the turn to avoid sharp turns and avoid sticking to the inside of the bend.
15. When collecting points at the corner of the wall, do not let the LiDar "face the wall", which will block the whole view. It is recommended to face away from the corner of the wall.
16. When collecting, it is recommended to release the device steadily and click the collection after 1~2s to avoid collection errors caused by delay.
17. In scenarios where GNSS is limited or without a signal, such as under eaves, behind buildings, or indoors, repeated initializations should be conducted if possible. First, conduct SLAM modeling in areas with stable GNSS coverage, focusing on the target to be measured. Once the environmental map is complete, slowly advance towards the target point. When reaching the target point, the laser head should move beyond the

previously scanned area to ensure accuracy. (Similar to RTK observations, having more satellites can improve positioning accuracy).

2.3.2 Installation environment

Operating environment: Hi-Survey software needs to run on the accompanying controller device, tablet, or other devices, which may cause compatibility issues such as crashes. You can copy the APK file (APK stands for Android Package) directly to the controller device or tablet for installation, or install it using a third-party mobile assistant on your computer.

Uninstall: Long press the Hi-Survey software to the top of the screen for uninstallation. The screen will pop up a window asking "Do you want to uninstall the software?" Click OK to uninstall the software.

Chapter 3

Equipment

This chapter contains:

- Device connection
- Base
- Rover
- Additional function
- Static collection
- Device information
- Data debugging
- Device check

3.1 Device connection

The controller can connect to the receiver via Bluetooth, network, WiFi, or NMEA universal devices (Bluetooth). Additionally, the controller can be set to have built-in GPS and a demo mode. To connect, open the Hi-Survey software, click on the [Please connect device] option in the upper right corner of the device menu, then enter the [Device] interface. Choose the connection method; it is recommended to use WiFi to connect to the host. Click [Connect], select the device you want to connect to (using the 'Device SN number'), and enter the WiFi password for the device. Enter the default password 12345678, and after entering the password, click [Confirm] to complete the connection.

If you choose the connection method as 'Bluetooth,' both the receiver and the controller must have their Bluetooth functions enabled. Click [Connect] in the lower right corner to enter the Bluetooth connection interface. Click [Search device] to find the device you want to connect to, select the instrument number of the receiver from the list, and a Bluetooth pairing dialog box will appear. Click 'Pair.' Once the Bluetooth pairing is successful, connect to the receiver. If you cannot find the device, click [Search device] below to search for the receiver again. After finding the corresponding instrument number, select the device and proceed with the connection.



Figure 3-1-1 Connect the device via WiFi

3.1.1 Register

Register the receiver, connect the GNSS receiver, and input the receiver registration code or scan the QR code of the receiver (please ask the relevant business personnel of Hi-Target for the registration code or QR code. The registration code of the receiver with 7-bit fuselage number is 21 bits, and the registration code of the receiver with 8-bit fuselage number is 24 bits).



Figure 3-1-2 Host registration

3.1.2 NFC connection

It is easy for you to use the NFC-enabled controller to scan the NFC tag of the receiver with NFC function, so as to connect the target device.

After the host is successfully connected, the device connection interface displays the basic information of the host, including working mode, firmware version, expiration time, host number, and automatically adapted antenna type. The connection button shows disconnect.

In the software setting interface, you can select NFC connection mode as "Bluetooth first" or "WiFi first". The default connection mode is Bluetooth.

3.2 Base

Setting the base station mainly sets the working parameters of the base station, including the coordinate of the base station, the data link of the base station and other parameters.

3.2.1 Template

You can save all the parameters set by the base station as a template, and then you can directly load the parameter template from the station setting template.

Long press to select the template, you can move or delete the template data.



Figure 3-2-1

[New] Click to enter the manual setting interface, you can configure the station parameters and save the new template.

[Load] Click on any template to jump to the manual site setting interface and load the site setting parameters of the template.

[Share] After logging in to the joint operation, click the share button to generate a 4-digit extraction code. Within half an hour, other operators can obtain the current station setting parameters based on the extraction code.

[Get] After logging in to the joint operation, click the [Get] button and enter the 4-digit extraction code to obtain the station setting template corresponding to the extraction code.

3.2.2 Set manually

You can select different data links to set the host.



Figure 3-2-2

[Save] Enter the template name to save the current parameters as a template.

[Generate] Generate the current parameter setting  as a QR code.

[Share] After logging in to the joint operation, click the [Share] to generate a 4-digit extraction code. Within half an hour, other operators can obtain the current station setting parameters based on the extraction code.

[Get] After logging in to the joint operation, click the [Get] and enter the 4-digit extraction code to obtain the station setting parameters corresponding to the extraction code.

3.2.3 Station location

Set the coordinates of the base station to the latitude and longitude in the WGS-84 coordinate system (Note: The H in the base station coordinates represents the ellipsoidal height. Since the main board requires the elevation height based on its internal model, you need to first obtain the elevation anomaly value of the elevation model at that location. This means setting the base station with GNSS in a measurable state. Therefore, when setting the base station and it prompts 'No elevation anomaly value obtained!', the current receiver's output solution status is unsolvable).



Figure 3-2-3

1. Set by average

Generally, when setting up the base station, you can collect a relatively accurate WGS-84 coordinate by averaging where the station is set up, so as to set up the station (Note: setting up the station at any position does not mean inputting any coordinates. It is necessary to set up the station after averaging for many times, and the more averaging times, the higher the reliability).

[Times] After selecting 'Set by average', the number of averaging input box is displayed. The average value is calculated by single-point positioning, and the number of averaging is set to 5 by default. In the new version, the information of each averaging is no longer displayed.

[Name] Name of the ground point where the base station is set.

[Target H] Input the instrument height and instrument height type of the base station.

2. Set by point

At known points, you can check 'Set by point', and input the local plane coordinates of the known point, or click the  button on the right to obtain it from the point library.

[Position] The ground point for setting the base station can be set manually or obtained through the point library.

[] Used to extract points from the coordinate library to the current interface (all point library buttons have similar functions).

Base station data link

Used to set the communication mode and parameters between the base station and the rover station, including 'Internal UHF' and 'Internal GSM'.



Figure 3-2-4

3. Internal UHF

When the base station uses the internal UHF function: set the data link as internal UHF, set the channel, radio protocol and power.



Figure 3-2-5

[Power] There are three options: High, Middle and Low.

[Frequency List] Entering the frequency table interface can obtain the optimal channel and set the frequency of a certain channel.



Figure 3-2-6

[Refresh]: If the current search results are not the best channel, you can change the starting channel to continue a new round of search.

[Modify]: Click on the frequency of a channel to modify the frequency value of the channel within the specified range.

[Save]: After the modification of the frequency table, you must click [Save]. The page prompts "Save frequency table success" to complete the modification of the frequency table.

[Reset]: Click OK to restore the frequency table to the default value.

Notice:



1. iRTK2/iRTK3 series (non-satel station) has no frequency table option;
2. The receiver of the iRTK platform and previous platforms (non-satel radio) can enter the ordinary frequency table interface and obtain the optimal channel and frequency table;
3. As long as the receiver with built-in Satel radio is available, it can enter another frequency table option interface to read and modify the radio protocol and channel;
4. A10 There is no frequency table option for radio stations other than XDL/ADL and satel;
5. The protocol selects Satel to set the radio FEC function;
6. When connecting to a host that does not support frequency point editing, you can view the frequency point but cannot edit it.

4. Internal GSM

When the base station uses the Internal GSM function: set the data link to the Internal GSM, set the Server/IP/port, and set the Group Type (By City Number, By Base Station SN).

The screenshot shows the 'Manual Setting' screen in the 'Start User' application. The interface includes a top status bar with system icons, a title bar with 'Start User' and 'Manual Setting', and a main content area with various settings. The 'Initial Job' section has two radio buttons: 'Set by average' (selected) and 'Set by point'. Below this is a 'Save Price' button with a play icon. The 'Name' field contains '000101'. The 'Target W' field contains '1.0000' with a 'Verify' button. The 'Connection Type' is set to 'Wired (LAN)'. The 'Resolution Mode' is set to '201'. The 'Server' field contains '0.000' with a 'Select' button. The bottom navigation bar has three icons: 'Save', 'Generate', and 'More'.

Figure 3-2-7

[Server]: Select the server, manually input the IP and Port number, or click [Select] to display the list of server addresses. You can select the required server from the list. At present, Hi-Target has multiple network servers and server ports for you to use, and you can choose the server and port by yourself.

Figure 3-2-8

[Group Type]: There are two types: By City Number and By Base Station SN.



Figure 3-2-9

[By City Number]: 7 digits and 3 digits respectively. The Group ID should be less than 255, and the base station and rover station need to be set in the same way to work properly.

[By Base Station SN]: The software automatically fills in the base station SN code; when synchronizing the rover station, you need to select the base station SN code mode and input the same base station SN code.

When the server is set to CORS mode, the base station logs into the server in CORS mode. You only need to enter the IP address and port number. The source node, username, and password are automatically set to the device ID and zhdgps, respectively, and cannot be changed. When setting up the rover station, the username should be an 8-digit number (it is recommended to use the device ID as the username for login), but it must not be the same as the base station's username. All other settings are the same as those for the base station.

3.3 Rover

Setting the rover station mainly sets the working parameters of the rover station, including the data link of the rover station and other parameters. The setting of the rover station is similar to that of the base station, but the input information is different.

3.3.1 Template

You can save all the parameters set by the rover station as a profile template, or you can load the parameters directly from the setting template.



Figure 3-3-1

3.3.2 Set manually

The method for manually setting up a rover station is the same as setting up a base station. When setting up a rover station, you can also choose to obtain configuration parameters by scanning a QR code. Typically, when using this method, the configuration parameter QR code is generated when setting up the base station. After sharing the QR code, users at the rover station can directly scan it to quickly obtain the parameters set for the base station and set up the station using these parameters, without the need for manual input.



Figure 3-3-2

3.3.3 Rover station data link

Used to set the communication mode and parameters between the rover station and the base station, including 'Internal UHF', 'Internal GSM', and 'Data Collector Internet'.



Figure 3-3-3

Rover stations use the 'Internal UHF' function: set the data link as a Internal UHF, modify the radio channel, radio protocol. The radio channel must be consistent with the base station.

Rover stations use the 'Internal GSM' function: The data link selects the Internal GSM, and you can click [Advance Config] to set other parameters such as network type.



Figure 3-3-4

[Server]: includes ZHD, CORS and TCP/IP. If you use Hi-Target server, use ZHD; if you access the CORS network, select CORS.

[IP]: manually enter the server IP and port number, or click [Select] to extract it. You can select the required server from the list.



Figure 3-3-5

[Group Type]: Select group number, By City Number or By Base Station SN.

[By City Number]: 7 digits and 3 digits respectively. The Group ID should be less than 255, and the base station and rover station need to be set in the same way to work properly.

[By Base Station SN]: When synchronizing the rover station, you need to select the base station SN code mode and input the same base station SN code.

[APN]: Enter 'CMNET' when using GPRS; enter 'card, card' when using CDMA.

[CORS]: Select CORS on the network and enter the IP address and port number of CORS.

Click the server, select CORS mode, click the [Set] button on the right side of the Mountpoint, and pop up the [CORS Parameters] interface. Click [Get] to obtain the CORS source list, select one node from the list, input [User Name] and [Password], and select the differential message format.

Notice:



In the CORS connection interface, there is a CORS node with RTCM correction. Check the RTCM1021-RTCM1027 option below, and then compare the actual coordinates given by the user with the output coordinates to ensure that the correction is correct.



Figure 3-3-6

[Open] You can invoke an existing CORS source node file (.txt).

[Save List] Generate the source node file (.txt) of the current CORS source node and save it in the folder which you want to save.

[OK] Complete the Settings and return to the previous screen.

The rover station uses the 'Data Collector Internet' function: in the rover station setting interface, select the data link as 'Data Collector Internet', and select the Server, set IP, Port, Mountpoint, User Name and Password, click [Set] to complete the rover setting.

Notice:



1. When the data link mode is 'Internal GSM', the APN parameter set by the operator is the parameter of the host SIM card;
2. When the data link mode is 'Data Collector Internet', the APN is set to the APN parameter of the controller system. In the case that the manual is not connected to WiFi, the manual must install a SIM card to set the APN, otherwise a prompt will be displayed;
3. For details on APN parameter setting in the manual system, refer to Manual and Application → APN parameter setting.

3.3.4 Other options for rover

Set the positioning data frequency, GNSS Elevation Mask, PPK and source node.

[Elevation Mask]: The cut-off Angle of the GNSS receiving satellite, which can be adjusted between 0 and 30 degrees.

[Pos Frequency]: After setting the output position data of positioning data frequency in the 'Advance Config' interface, in the detailed measurement interface, if you select the option to automatically collect data by time and check the synchronization of positioning data output, it will automatically synchronize with the positioning data frequency. For example, if the positioning data frequency is 2Hz, the data will be transmitted every 0.5 seconds.

[Store GNSS Raw]: After this function is enabled in Rover, the receiver synchronization starts temporary static collection. In 'Detail Survey', click the floating collection button to perform PPK collection operation and record the RSP file. The file name of the file is consistent with that of the ongoing static collection.

[Set]: Generally, click after all the rover station parameters are set. A prompt box will pop up after clicking. If the setting is successful, check whether the rover can normally receive the differential signal. If the setting fails, check whether the parameters are set incorrectly and click several times repeatedly.

3.4 Additional function

The receiver auxiliary functions include: module information, registration information, satellite tracking switch, receiver setting, service information, restore system, network diagnosis, location service and Data Transmission.

3.4.1 Module information

View the type and status information of the communication module, as well as the firmware version number. This view of the module information is only supported for some models.

'Radio Module Type': Displays the type information of the main radio module.

'Radio firmware version': Displays the firmware version of the main radio module.

'Network Module Type': Displays the type information of the current network module.

'IMU Module Type': Displays the type information of the IMU module on the host.

'IMU firmware version': Displays the firmware version of the IMU module on the host.



Figure 3-4-1

3.4.2 Registration

Under the registration information interface, view the QR code, registration code and expiration time of the instrument registration information.



Figure 3-4-2

3.4.3 Satellite tracking switch

Enter "Satellite Tracking Switch" to select whether the satellite system is on or off. When the satellite system is turned off, it will not be displayed in the satellite map.

[Note]: All satellite systems will be opened when static recording begins.



Figure 3-4-3

3.4.4 Receiver settings

Set the receiver's static RINEX record, firmware update prompt, sound type, voice volume, Hi-Fix, SLAM Laser Echo Num, SLAM Laser Frame Rate.



Figure 3-4-4

[Store RINEX Data]: If checked, Rinex format data will be recorded synchronously when static data is collected (current firmware version does not support this feature).

[Firmware Update Prompt]: When enabled, a prompt is displayed when there is firmware that can be upgraded.

[Sound Type]: Switches voice announcements (none/default/customized (in English mode)).

[Volume]: The volume of the host voice can be set.

3.4.5 Service information

Displays the current receiver version type and its corresponding function permissions.



Figure 3-4-5

3.4.6 Restore System

When you are connected to a host, you can restore the firmware upgrade series to its original state.

3.4.7 Network Diagnosis

This function checks whether the network module and SIM card are functioning properly, the strength of the network signal, whether  they are registered  to the network,  whether dialing is successful, and whether a successful connection to the server has been established. It indicates that this step is normal; it indicates that this step is abnormal; it indicates that this process has not been checked (default state).

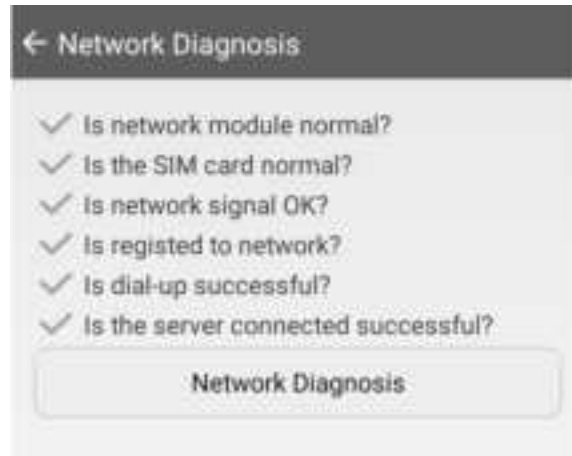


Figure 3-4-6

3.4.8 Data Transmission

This function can forward the differential data of the base station to other devices within the same LAN, so that other devices can set up the differential rover station to obtain the differential. When connecting to a host that supports the differential base station controller, this entry is displayed.



Figure 3-4-7

1. Environmental preparation

- Multiple controllers need to be connected to the same LAN;
- The controller 1 is connected to a host that supports the base station controller differential data link;
- The remaining controllers connect to the host that supports the rover station controller differential data link.

2. operating process

- Enter the "Base Station Diff Data Transmission" interface in controller 1, set the station parameters, and open the switch to set the host as the base station differential CORS, and switch the interface to display the current IP address of the controller and the forwarding port.

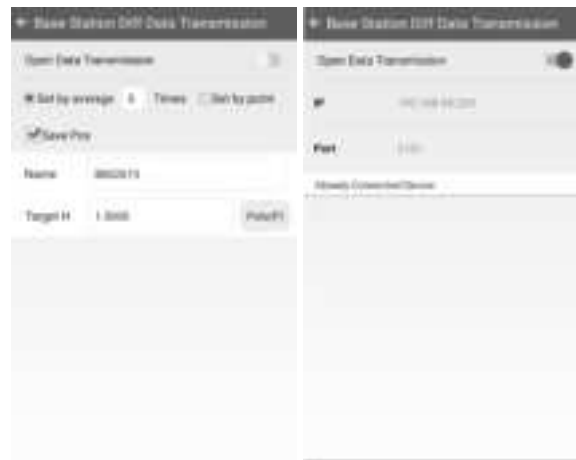


Figure 3-4-8

- Set the host of the rover station's controller to the rover station's controller differential CORS. Enter the IP port displayed on the "Base Station Diff Data Transmission" interface, the source node input the base station SN number or RTCM32, and enter the user name and password (the default password is zhdgps) to obtain the transmitted differential data and achieve fixation.

3.5 Static collection

3.5.1 Static collection Settings

In RTK working mode, if static collection is required at the same time, you can click the static collection menu to enter the temporary static collection Settings, input the interval, file name, pole height and elevation mask; you can view GDOP, file size, start time and record time. After clicking the "Start" button, the recording starts.

If the current connected receiver is in the A10/V60/iRTK2 series or iRTK5 series, you can set the [Set Duration] and select the [Static Mode]. After selecting the [Static Mode], the [Set Duration] function will be enabled. Once the specified collection duration is reached, the collection will stop, and the receiver will automatically turn off. To cancel the [Static Mode] option, you must first set the receiver to either a reference station or a rover station. After unselecting, the static collection will stop.



Figure 3-5-1

3.5.2 Static file management

You can view the static files in the current receiver and perform related operations.



Figure 3-5-2

[Refresh] Refresh the current file list interface.

[Delete] Long press a record to delete the selected static data, support multiple files for deletion.

[Format] Format static data, the data is not recoverable.

When connecting to a WiFi network, the receiver supports FTP downloads of static files to the local storage of the device. Long-pressing a static file will bring up a file selection mode (multiple files can be selected). After a successful download, the software will prompt you with the file download and save path. If the user is not connected to WiFi, clicking the download button will result in a message stating that 'This operation must be working in Wi-Fi connection model.



ID	File Name	Size	Date
1	HT10001.DAT	100.0 KB	2023-09-12
2	HT10002.DAT	100.0 KB	2023-09-12
3	HT10003.DAT	100.0 KB	2023-09-12
4	HT10004.DAT	100.0 KB	2023-09-12
5	HT10005.DAT	100.0 KB	2023-09-12
6	HT10006.DAT	100.0 KB	2023-09-12
7	HT10007.DAT	100.0 KB	2023-09-12
8	HT10008.DAT	100.0 KB	2023-09-12

Figure 3-5-3

3.6 Device information

The device information interface displays the device information, module information and service authorization information of the current connected host.

The host device information includes: SN number, Receiver firmware version, motherboard version, expiration time, working mode;

Module information includes: radio module type, radio firmware version, network module type, IMU module type, IMU firmware version;

Service authorization information: motherboard serial number, Hi-Target service expiration time and other content.



Device Information	
SN	HT1000000
Receiver FW	1.0.0.0 V1.0.0
Motherboard FW	1.0.0.0 (1.0)
Expiration	2023-09-12
Working Mode	Basic Mode
Radio Module Type	HT-RTK-TS4400-B
Radio Firmware Version	V1.0.0
Network Module Type	4G/5G
IMU Module Type	SC05A34
IMU Firmware Version	V1.0.0
Board SN	HT10000000000

Figure 3-6-1

3.7 Data debugging

It is mainly used to debug data and detect GPRS signal strength. The received debugging data can be saved as a file.

[Hex] Selects and displays in hexadecimal format.

[Refresh] Select and refresh output.

[Save] Save the output data.

[New Line] Select, send new line. In general, you must select the command before sending it.

[Send] Enter the command above and click the "send" command.



Figure 3-7-1

Horizontal sensor data output is added in demo mode and built-in GPS mode.



Figure 3-7-2

3.8 Device check

Device check is the latest function of this software, that is, to check whether the coordinate point meets the specified accuracy.

Under the main interface of the device, click the [Device Check] icon to enter the device check interface. Users can fill in or select the corresponding coordinate points from the known points, and the software will display the corresponding NEZ. Then, click the [Start Check] button, and the software will process the data, displaying it in a list at the bottom of the interface. The displayed content includes: Point Name N, Difference N, E, Difference E, Z, Difference Z. After the device check is successful, the software will prompt with a message saying "Current equipment meets inspection"



Figure 3-8-1

Chapter 4

Measure

This chapter contains:

- Detail survey
- Tilt measurement
- SLAM-RTK mode
- SLAM image measurement mode
- SLAM
- Download of job data
- Device check
- Data processing

4.1 Detail survey

Click the [Fragment Measurement] button on the main menu of the measurement page to enter the fragment measurement interface. The text interface and graphic interface can be switched by clicking the [Text] / [Graphic] button.



Figure 4-1-1



Figure 4-1-2



Figure 4-1-3

All tool buttons can be configured to display or hide in [Measurement Configuration] -> [Display] -> "Tool Configuration".

4.1.1 Manual collection

In most cases, upon reaching the measurement location, the system determines whether to proceed with data collection based on the measurement  coordinates and their accuracy, as well as the solution status displayed on the interface. In RTK fixed solutions, when you click 'manual collection,' the software first performs an accuracy check (the accuracy settings are detailed in 'Measurement Configuration' -> 'Data'). If the accuracy does not meet the requirements, the software will provide a prompt.

Before the data collection is complete, a detailed interface for saving coordinate point information will appear, allowing you to check the reliability of the points. The software automatically accumulates the points based on their global point numbers, and the prefix of the point name reflects the historical usage. You can directly enter the 'target height' or click 'pole height' to configure the target height and antenna type in detail. In the 'legend

description' section, you can input annotation information or select common annotation types. Additionally, you can set the distance in the collection confirmation box.



Figure 4-1-4

Figure 4-1-5

4.1.2 Automatic acquisition

Click the automatic  acquisition button to enter the automatic acquisition settings interface. Select the automatic acquisition mode, including time interval, planar interval (change in N or E direction), oblique distance interval, and electronic bubble centering. Enter the prefix, number, and annotation information for the point. After clicking [OK], the software will enter the automatic acquisition mode. During the automatic acquisition process, accuracy checks will be performed (the accuracy  settings are detailed in 'Measurement Configuration' → 'Data'). If the accuracy requirements are met, the software will automatically acquire and prompt you to save the point. If the accuracy requirements are not met, no prompts will be given until the accuracy is satisfied, at which point the software will continue to acquire and prompt you to save the point. Click to end the automatic acquisition.



Figure 4-1-6



Figure 4-1-7

When the automatic acquisition mode is selected as "time", the step length (s) can be set, and the acquisition will be automatically carried out according to the set time interval.

When the automatic acquisition mode is selected as "by flat distance", the step length value can be set, and the acquisition will be automatically carried out according to the flat distance step length value.

When the automatic acquisition mode is selected as "acquisition according to the slope distance", the step length value can be set, and the acquisition will be automatically acquired according to the slope distance step length value.

When the automatic acquisition mode is selected as "electronic bubble centering", you can "straighten and collect, and leave after collecting", without manual intervention. Click [Configuration] - [Data] - [Electronic Bubble Configuration] in the upper right corner of the page to set the bubble accuracy.

When the data quality is poor (the type of solution, the accuracy of coordinates, and the centering of bubbles do not meet the requirements), the automatic collection will be suspended, and specific red font prompts will be given according to the current unsatisfactory conditions.

4.1.3 Average collection

Average collection is a simple method to enhance measurement accuracy. According to error theory, errors can occur in any direction. Therefore, with a sufficient number of observations, random errors will cancel each other out (though this is only theoretical; in practice, more smoothing does not necessarily mean higher

accuracy). After entering the smoothing interface and clicking start, the software begins to plot coordinates and display the current position. During the smoothing process, the software automatically analyzes the data quality, calculates the standard deviation (median error), and displays it.

Smooth acquisition supports both text and graphical displays. Once the smooth acquisition has reached a certain number of times, you can remove specific points from the smooth acquisition list in the text mode interface to improve accuracy by not including these points in the smoothing process (long press the smooth acquisition item to see the 'Delete' toolbar). The software will automatically recalculate the smoothness. In the graphical display interface, clicking on a point from the smooth acquisition will show its coordinates.



Figure 4-1-8

Figure 4-1-9

The calculated median error is theoretically the same as the measured median error; however, due to the smoothing process being a small sample collection, the estimated median error may be less than the actual measurement error.

Before performing smooth data collection, you can configure the smoothing settings by clicking on [Configuration] in the upper right corner of the collection interface. This allows you to set the smoothing method, type of solution, number of smoothings, and smoothing accuracy. Each time you exit the software, the solution type for smooth data collection will be set to fixed. During smooth data collection, no further configurations can be made. If the current solution type does not match the set one, the software will display a red alert stating "Data solution status does not meet the requirements, smooth paused." In this case, you need to reset the solution type. If the smoothness accuracy is not met or the configuration exceeds the limit, the software will display a warning "Data quality is poor, smooth paused." In this case, you need to reconfigure the smoothness accuracy.



Figure 4-1-10



Figure 4-1-11

4.1.4 PPK collection

PPK (GPS dynamic acquisition post-processing differential) technology is the earliest GPS dynamic differential technology, also known as stop and go.

The mobile station uses the PPK acquisition function. After filling in the parameters of the mobile station, expand the advanced options below the station setting interface and open the "PPK" switch. Click the setting to set the station. The mobile station starts recording static data, enters the graphic acquisition interface for detailed measurement, and clicks the floating collection button to enter the PPK acquisition interface.



Figure 4-1-12

Figure 4-1-13

Figure 4-1-14

Enter the corresponding point name, target height and legend description at the measurement position, select the number of times to be smoothed (generally recommended more than 200 times), and click [Start marking] to collect.

When the set number of smooth times is reached or click [End Mark], the software will automatically save the coordinate points.

Each PPK collection will correspond to the recording of a PPK file record, which can be viewed and deleted in the PPK file management module.

Notice:


The data and marking information of the PPK function of the mobile station are recorded in the Rinex folder of the host in Rinex format, and the Rinex file should be used for calculation during post-processing. For the relevant operations of PPK post-processing, please refer to the "HGO V2.0.0 Software User Manual".

4.2 General Measurement

4.2.1 Tilt measurement

After connecting the host to Hi-Survey measurement software, open the "tilt measurement" switch in the measurement interface or [Measurement Configuration] -- [Data] interface. Click the tilt measurement switch in the measurement interface or the tilt measurement icon in the top message bar to trigger the initialization pop-up window. Perform the initialization of tilt measurement according to the prompt in the pop-up window.

According to the prompt of the initialization pop-up window, shake the RTK host back and forth (frequency is 1 second) for heading initialization;



Figure 4-2-1



Figure 4-2-2



Figure 4-2-3



Figure 4-2-4

When all the initialization items are checked, the host voice announces "enter tilt measurement", and the tilt measurement icon is lit. After the tilt measurement initialization is completed, the host can perform the tilt measurement operation.

Notice:



1. After the tilt measurement is enabled, the uncorrected tilt measurement must be initialized before it can be used. Otherwise, the user will be prompted to perform initialization operation.
2. After the uncorrected tilt measurement is turned on, click the icon of tilt measurement in the lower right corner of the interface to select the off tilt measurement.
3. Start the tilt measurement and enable the tilt measurement mode. The bottom box can view the tilt azimuth and the page that has not been corrected for tilt.
4. After entering the uncorrected tilt measurement, when the electronic bubble switch is turned on, the vertical tilt Angle is displayed above the electronic bubble, and when the electronic bubble switch is turned off, the vertical tilt Angle is displayed in the bottom box.

4.2.2 SLAM-RTK mode

Click SLAM on the left toolbar, select SLAM-RTK mode, and complete the initialization according to the software prompts. Then enter SLAM-RTK mode. The state of SLAM is switched to SLAM solution state, and the coordinates are displayed as SLAM trajectory coordinates.



Figure 4-2-5



Figure 4-2-6



Figure 4-2-7



Figure 4-2-8



Figure 4-2-9

4.2.3 SLAM image measurement mode

Click on the SLAM icon in the toolbar, select the SLAM image measurement mode, and navigate to the puncture point interface. Follow the software prompts to complete the initialization. After initialization, switch to the SLAM solution state, where the coordinates will display as SLAM trajectory coordinates. Click the 'Shooting' button to take a single photo. Move the pointer to select the desired puncture point position, click 'OK', then move the pointer again to select another position on the same plane as the puncture point. Click 'Save' to calculate and save the coordinates of the puncture point. Once saved, click the 'Point Library' button to view the point data and coordinate library.